

JavaScript Roadmap (July → August 17)

Goal: Build a strong foundation in modern JavaScript in approximately **100 hours** (3 hours/day) before August 17.

📌 Objectives

By the end of this roadmap, I should be able to:

- Build applications without following tutorials step-by-step.
 - Understand how JavaScript works under the hood.
 - Manipulate the DOM confidently.
 - Consume REST APIs.
 - Use asynchronous programming.
 - Organize projects using ES Modules.
 - Create projects worthy of my GitHub portfolio.
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📌 Main Resources

📌 Primary Resource

- <https://javascript.info/>

This will be my main textbook.

📌 Secondary Resource

Eloquent JavaScript (3rd Edition)

<https://eloquentjavascript.net/>

Read only selected chapters.

📌 Deep Understanding

You Don't Know JS Yet

<https://github.com/getify/You-Dont-Know-JS>

Focus on:

- Get Started
 - Scope & Closures
 - Async & Performance (selected chapters)
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📌 Reference

MDN Documentation

<https://developer.mozilla.org/>

Use it whenever I encounter a new API or method.

📌 Practice

- Codewars
- LeetCode (Easy)

Goal:

- 2–3 problems every day
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📅 Daily Schedule (3 Hours)

Time	Activity
1h	Read theory
1h15min	Coding exercises
45min	Weekly project

Week 1 — JavaScript Fundamentals

Goal

Become comfortable writing JavaScript.

Topics

- Variables (`let` , `const`)
 - Primitive Types
 - Operators
 - Strings
 - Template Literals
 - Conditionals
 - Loops
 - Functions
 - Arrow Functions
 - Scope
 - Arrays
 - Objects
-

Read

javascript.info

- JavaScript Fundamentals
- Code Quality
- Objects: The Basics

Eloquent JavaScript

- Chapter 1
 - Chapter 2
 - Chapter 3
 - Chapter 4
-

Mini Projects

- Guess the Number
 - Calculator
 - Temperature Converter
 - BMI Calculator
 - Password Validator
 - Palindrome Checker
 - Prime Number Checker
 - Rock Paper Scissors
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End of Week Checklist

Knowledge

- ☐ I understand variables and primitive types.
- ☐ I can write loops without consulting notes.
- ☐ I understand the difference between `let` and `const`.
- ☐ I can create and manipulate arrays.
- ☐ I can create and manipulate objects.
- ☐ I can write functions comfortably.

Practice

- ☐ I completed at least **6 mini projects**.
- ☐ I solved **10+ coding exercises**.

Self Rating

Knowledge:

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Confidence:

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Week 2 — Thinking Like a JavaScript Developer

Goal

Learn what makes JavaScript different from other languages.

Topics

- Object references
- Destructuring
- Spread Operator
- Rest Parameters
- Array Methods

Especially:

- `map()`
- `filter()`
- `reduce()`
- `find()`
- `every()`
- `some()`
- `sort()`

Then:

- Higher-order Functions
- Closures
- Lexical Scope
- `this`

Read

- Data Types
- Advanced Functions
- Object Properties

You Don't Know JS

- Get Started
- Scope & Closures

Weekly Project

Expense Tracker

Features

- Add expense
- Delete expense
- Categories
- Total balance
- Monthly total
- Filter by category

(No LocalStorage yet.)

End of Week Checklist

Knowledge

- ☐ I understand higher-order functions.
- ☐ I know when to use map/filter/reduce.
- ☐ I understand closures.
- ☐ I understand lexical scope.
- ☐ I understand basic usage of `this`.

Practice

- ☐ Expense Tracker completed.
- ☐ I solved 15+ coding problems.

Self Rating

Knowledge:

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Confidence:

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Goal

Start building real applications.

Topics

- DOM
 - `querySelector()`
 - `createElement()`
 - `appendChild()`
 - `remove()`
 - Events
 - Event Delegation
 - Forms
 - LocalStorage
 - JSON
 - Fetch API
-

Read

javascript.info

- Document
 - Browser
 - UI Events
 - Network Requests
-

Project 1

To-do List

Features

- Add task
 - Delete task
 - Edit task
 - Complete task
 - LocalStorage
 - Filter completed
-

Project 2

Weather App

Using:

- OpenWeather API
- or WeatherAPI

Display:

- Temperature
 - Humidity
 - Wind
 - Weather Icon
-

End of Week Checklist

Knowledge

- ☐ I understand the DOM.

- ☐ I can manipulate HTML using JavaScript.
- ☐ I understand events.
- ☐ I can use LocalStorage.
- ☐ I can consume APIs using Fetch.

Practice

- ☐ To-do List completed.
- ☐ Weather App completed.

Self Rating

Knowledge:

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Confidence:

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Week 4 — Modern JavaScript

Goal

Learn modern JavaScript features used in professional projects.

Topics

- ES Modules
- import/export
- Classes
- Prototypes (basic understanding)
- Promises
- Async/Await
- try/catch
- Error handling

Read

javascript.info

- Modules
- Promises
- Async/Await

You Don't Know JS

- Async & Performance (selected chapters)

Weekly Project

Movie Search App

Using TMDB API.

Features

- Search movies
- Pagination
- Favorites
- Loading spinner
- Error handling

End of Week Checklist

Knowledge

- ☐ I understand Promises.
- ☐ I understand Async/Await.
- ☐ I know how modules work.
- ☐ I understand basic classes.
- ☐ I know how to handle errors.

Practice

- ☐ Movie Search App completed.

Self Rating

Knowledge:

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Confidence:

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Final Project (August 12 → August 17)

Choose ONE.

Option 1

Habit Tracker

Features

- Streaks
- Categories
- Daily progress
- LocalStorage

Option 2

Kanban Board

Features

- Drag & Drop
- Create tasks
- Edit tasks
- Delete tasks

Option 3 (Recommended)

Notes App

Features

- Markdown Preview
- Search
- Categories
- Sorting
- LocalStorage
- Dark Mode

Portfolio Checklist

By August 17 I should have:

- ☐ Expense Tracker
- ☐ To-do List
- ☐ Weather App
- ☐ Movie Search App
- ☐ Final Project
- ☐ GitHub repository updated
- ☐ README written for every project

Daily Habit Tracker

Habit	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Studied 3h	☐	☐	☐	☐	☐	☐	☐
Read theory	☐	☐	☐	☐	☐	☐	☐
Solved exercises	☐	☐	☐	☐	☐	☐	☐
Worked on project	☐	☐	☐	☐	☐	☐	☐
Code pushed to GitHub	☐	☐	☐	☐	☐	☐	☐

Final Self-Evaluation

JavaScript

- ☐ Variables
- ☐ Functions
- ☐ Objects
- ☐ Arrays
- ☐ Closures
- ☐ Scope
- ☐ DOM
- ☐ Events
- ☐ Fetch API
- ☐ Promises

- ☐ Async/Await
 - ☐ LocalStorage
 - ☐ Modules
 - ☐ Error Handling
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Confidence Level

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Next Step After August 17

- Learn TypeScript
- Learn npm
- Learn Vite
- Learn React
- Build larger portfolio projects
- Learn Node.js and Express